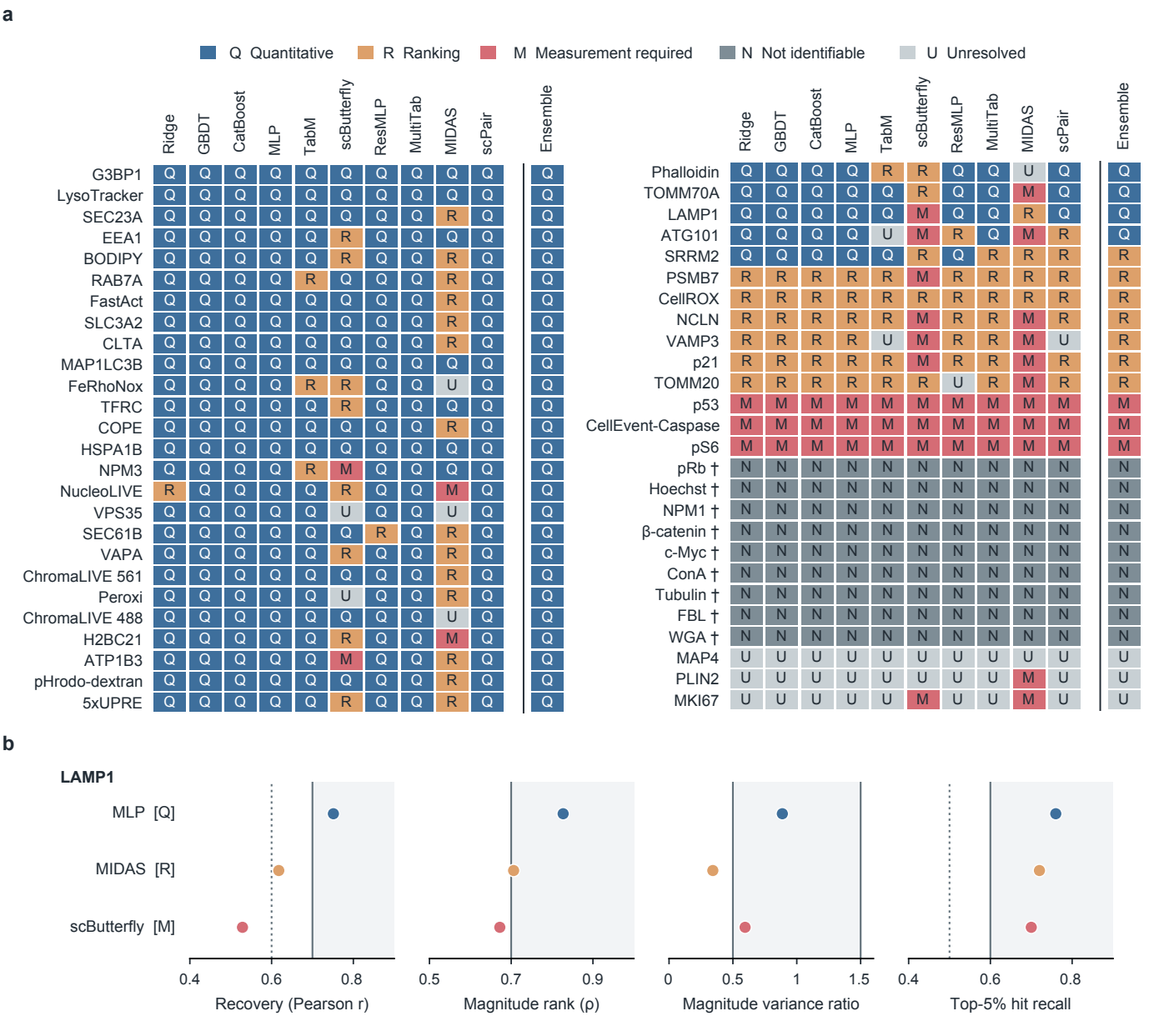

a
Q Quantitative
R Ranking
M Measurement required
N Not identifiable
U Unresolved
Ridge
GBDT
CatBoost
MLP
TabM
scButterfly
ResMLP
MultiTab
MIDAS
scPair
Ensemble
G3BP1 Q Q Q Q Q Q Q Q Q Q Q
LysoTracker Q Q Q Q Q Q Q Q Q Q Q
SEC23A Q Q Q Q Q Q Q Q R Q Q
EEA1 Q Q Q Q Q R Q Q Q Q Q
BODIPY Q Q Q Q Q R Q Q R Q Q
RAB7A Q Q Q Q R Q Q Q R Q Q
FastAct Q Q Q Q Q Q Q Q R Q Q
SLC3A2 Q Q Q Q Q Q Q Q R Q Q
CLTA Q Q Q Q Q Q Q Q R Q Q
MAP1LC3B Q Q Q Q Q Q Q Q Q Q Q
FeRhoNox Q Q Q Q R R Q Q U Q Q
TFRC Q Q Q Q Q R Q Q Q Q Q
COPE Q Q Q Q Q Q Q Q R Q Q
HSPA1B Q Q Q Q Q Q Q Q Q Q Q
NPM3 Q Q Q Q R M Q Q Q Q Q
NucleoLIVE R Q Q Q Q R Q Q M Q Q
VPS35 Q Q Q Q Q U Q Q U Q Q
SEC61B Q Q Q Q Q Q R Q R Q Q
VAPA Q Q Q Q Q R Q Q R Q Q
ChromaLIVE 561 Q Q Q Q Q Q Q Q R Q Q
Peroxi Q Q Q Q Q U Q Q R Q Q
ChromaLIVE 488 Q Q Q Q Q Q Q Q U Q Q
H2BC21 Q Q Q Q Q R Q Q M Q Q
ATP1B3 Q Q Q Q Q M Q Q R Q Q
pHrodo-dextran Q Q Q Q Q Q Q Q R Q Q
5xUPRE Q Q Q Q Q R Q Q R Q Q
Phalloidin Q Q Q Q R R Q Q U Q Q
TOMM70A Q Q Q Q Q R Q Q M Q Q
LAMP1 Q Q Q Q Q M Q Q R Q Q
ATG101 Q Q Q Q U M R Q M R Q
SRRM2 Q Q Q Q Q R Q R R R R
PSMB7 R R R R R M R R R R R
CellROX R R R R R R R R R R R
NCLN R R R R R M R R M R R
VAMP3 R R R R U M R R M U R
p21 R R R R R M R R M R R
TOMM20 R R R R R R U R M R R
p53 M M M M M M M M M M M
CellEvent-Caspase M M M M M M M M M M M
pS6 M M M M M M M M M M M
pRb † N N N N N N N N N N N
Hoechst † N N N N N N N N N N N
NPM1 † N N N N N N N N N N N
β-catenin † N N N N N N N N N N N
c-Myc † N N N N N N N N N N N
ConA † N N N N N N N N N N N
Tubulin † N N N N N N N N N N N
FBL † N N N N N N N N N N N
WGA † N N N N N N N N N N N
MAP4 U U U U U U U U U U U
PLIN2 U U U U U U U U M U U
MKI67 U U U U U M U U M U U
b
LAMP1
MLP [Q]
MIDAS [R]
scButterfly [M]
0.4
0.6
0.8
Recovery (Pearson r)
0.5
0.7
0.9
Magnitude rank (ρ)
0
0.5
1
1.5
Magnitude variance ratio
0.4
0.6
0.8
Top-5% hit recall